

Lecture 2

Binary Codes and Binary Logic

BCD and other decimal codes · Gray code · ASCII and parity · registers · AND, OR, NOT and timing diagrams

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 1, §1.7–1.9

P	b ₆	b ₅	b ₄	b ₃	b ₂	b ₁	b ₀
1	1	0	0	0	0	0	1

ASCII 'A' with odd parity

Today's roadmap

1

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Decimal codes

BCD, 2421, excess-3

2

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Gray, ASCII, parity

One-bit change; error check

3

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Registers

Storing and moving bits

4

§1.9

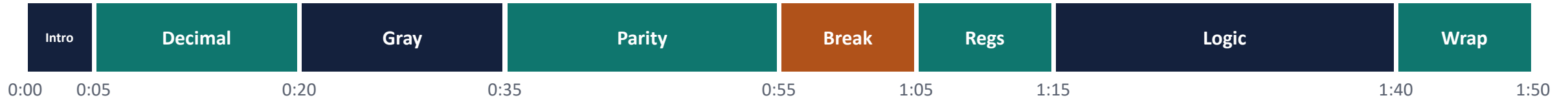
Binary logic

AND, OR, NOT, timing

By the end of class you can...

- Encode decimal digits in BCD, 2421 and excess-3
- Convert between binary and Gray code
- Generate and check even and odd parity
- Interpret a register's contents under a stated code
- Evaluate AND, OR, NOT and draw timing diagrams

Agenda for today's session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	—
0:05 – 0:20	BCD and other decimal codes	Example 1
0:20 – 0:35	Gray code	Example 2
0:35 – 0:55	ASCII and parity	Example 3 · Your turn 1
0:55 – 1:05	Break	
1:05 – 1:15	Binary storage and registers	Register example
1:15 – 1:40	AND, OR, NOT, signal levels, timing	Example 4 · Your turn 2
1:40 – 1:50	Concept check and wrap-up	Homework preview

Four codes for the decimal digits

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Digit	BCD 8421	2421	Excess-3	8 4 -2 -1
0	0000	0000	0011	0000
1	0001	0001	0100	0111
2	0010	0010	0101	0110
3	0011	0011	0110	0101
4	0100	0100	0111	0100
5	0101	1011	1000	1011
6	0110	1100	1001	1010
7	0111	1101	1010	1001
8	1000	1110	1011	1000
9	1001	1111	1100	1111

Each code leaves 6 of the 16 patterns unused.

Weighted codes

Digit = sum of weights of the 1 bits.

2421: $6 = 1100 = 2 + 4$

8 4 -2 -1: $6 = 1010 = 8 - 2$

Self-complementing

Inverting every bit of digit d gives the code of $9 - d$.

Excess-3: $3 = 0110 \rightarrow 1001 = 6$ ✓

BCD is not: $0011 \rightarrow 1100$, which is invalid.

Example 1 — Binary versus BCD

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EXAMPLE

Represent $(185)_{10}$ in straight binary and in BCD. Compare the bit counts.

Straight binary (8 bits)

128	64	32	16	8	4	2	1
1	0	1	1	1	0	0	1

BCD (12 bits): one group per digit

0	0	0	1	1	0	0	0	0	1	0	1
1				8				5			

Read as binary, the BCD pattern would be 389: the code sets the meaning.

- 1 Largest power of 2 in 185 is 128: $185 - 128 = 57$
- 2 $57 - 32 = 25$, $25 - 16 = 9$, $9 - 8 = 1$, $1 - 1 = 0$
- 3 1s at 128, 32, 16, 8, 1: **$(10111001)_2$** , 8 bits
- 4 BCD: split into decimal digits 1, 8, 5
- 5 Code each digit in 4 bits: $1 = 0001$, $8 = 1000$, $5 = 0101$
- 6 Join them: **$(0001\ 1000\ 0101)$** _BCD, 12 bits
- 7 BCD costs bits but drives displays with no conversion.

Gray code: one bit changes at a time

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Dec	Binary	Gray	Dec	Binary	Gray
0	0000	0000	8	1000	1100
1	0001	0001	9	1001	1101
2	0010	0011	10	1010	1111
3	0011	0010	11	1011	1110
4	0100	0110	12	1100	1010
5	0101	0111	13	1101	1011
6	0110	0101	14	1110	1001
7	0111	0100	15	1111	1000

Binary → Gray

$$g_{n-1} = b_{n-1}, \quad g_i = b_{i+1} \oplus b_i$$

Shortcut: XOR the word with itself shifted right one place.

Gray → binary

$$b_{n-1} = g_{n-1}, \quad b_i = b_{i+1} \oplus g_i$$

Sequential: each bit uses the binary bit just found.

Example 2 — Gray code both ways

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EXAMPLE

(a) Convert binary 1011 0110 to Gray. (b) Convert Gray 1110 1001 to binary.

(a) XOR neighbouring bits

binary	1	0	1	1	0	1	1	0
Gray	1	1	1	0	1	1	0	1

(b) XOR with the binary bit just found

Gray	1	1	1	0	1	0	0	1
binary	1	0	1	1	0	0	0	1

- 1 (a) Copy the leftmost bit: $g_7 = b_7 = 1$
- 2 XOR each pair of neighbours: $1 \oplus 0 = 1$, $0 \oplus 1 = 1$, $1 \oplus 1 = 0$
- 3 ... $1 \oplus 0 = 1$, $0 \oplus 1 = 1$, $1 \oplus 1 = 0$, $1 \oplus 0 = 1$
- 4 Gray = **1110 1101**
- 5 (b) Copy the leftmost bit: $b_7 = g_7 = 1$
- 6 XOR last binary bit with next Gray bit:
- 7 $1 \oplus 1 = 0$, $0 \oplus 1 = 1$, $1 \oplus 0 = 1$, $1 \oplus 1 = 0$...
- 8 ... $0 \oplus 0 = 0$, $0 \oplus 0 = 0$, $0 \oplus 1 = 1$
- 9 Binary = **1011 0001**

ASCII and the parity bit

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ASCII

7 bits, 128 characters: 94 printing, 34 control.

'0' = 011 0000 (48) 'A' = 100 0001 (65) 'a' = 110 0001 (97)

Parity bit

An 8th bit that makes the total number of 1s even (even parity) or odd (odd parity).

A single parity bit detects any odd number of bit errors and corrects none.

Character	Even parity	Odd parity
A = 100 0001	0100 0001	1100 0001
T = 101 0100	1101 0100	0101 0100

Generate

Count the 1s in the data. Even parity: $P = 1$ when the count is odd.

Check

Count all received bits including P. The wrong total flags an error: ask for the word again.

Example 3 — Checking parity

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EXAMPLE

ASCII 'A' is sent with EVEN parity. What is sent? The receiver gets (b) 1100 0001, (c) 1100 0011.

sent

0	1	0	0	0	0	0	1
---	---	---	---	---	---	---	---

two 1s + $P=0 \rightarrow$ even

(b)

1	1	0	0	0	0	0	1
---	---	---	---	---	---	---	---

three 1s $\rightarrow$ ERROR

(c)

1	1	0	0	0	0	1	1
---	---	---	---	---	---	---	---

four 1s $\rightarrow$ passes!

- 1 Count the 1s in the data 100 0001: two.
- 2 Two is already even, so $P = 0$. Sent: **0100 0001**
- 3 (b) Count the 1s in 1100 0001: three.
- 4 Three is odd: the error is **detected**.
- 5 The receiver cannot tell which bit is wrong: it asks again.
- 6 (c) Count the 1s in 1100 0011: four.
- 7 Four is even: the two-bit error is **missed**.

Your turn 1 — Codes and parity

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YOUR TURN

≈ 10 min

- (a) Write $(729)_{10}$ in BCD and in excess-3.
- (b) Write $(13)_{10}$ in 4-bit Gray code. What decimal value is the Gray word 0111?
- (c) Give the 8-bit word for ASCII 'T' = 101 0100 with even parity and with odd parity (parity bit on the left).

Check (b) by converting your Gray answer back.

Your turn 1 — Solution

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SOLUTION

(a) Digit by digit: 7, 2, 9

BCD	0	1	1	1	0	0	1	0	1	0	0	1
XS-3	1	0	1	0	0	1	0	1	1	1	0	0

(b) 13 = 1101 in binary

binary	1	1	0	1
Gray	1	0	1	1

- 1 (a) BCD: 7 = 0111, 2 = 0010, 9 = 1001
- 2 Excess-3 (digit + 3): 1010, 0101, 1100
- 3 (b) 13 = 8 + 4 + 1 = 1101
- 4 Gray: copy 1, then $1 \oplus 1 = 0$, $1 \oplus 0 = 1$, $0 \oplus 1 = 1$: **1011**
- 5 Gray 0111 → binary: 0, $0 \oplus 1 = 1$, $1 \oplus 1 = 0$, $0 \oplus 1 = 1$
- 6 Binary 0101 = 4 + 1 = **5**
- 7 (c) 'T' = 101 0100 has three 1s.
- 8 Even parity: add $P = 1 \rightarrow$ **1101 0100**
- 9 Odd parity: add $P = 0 \rightarrow$ **0101 0100**

Break

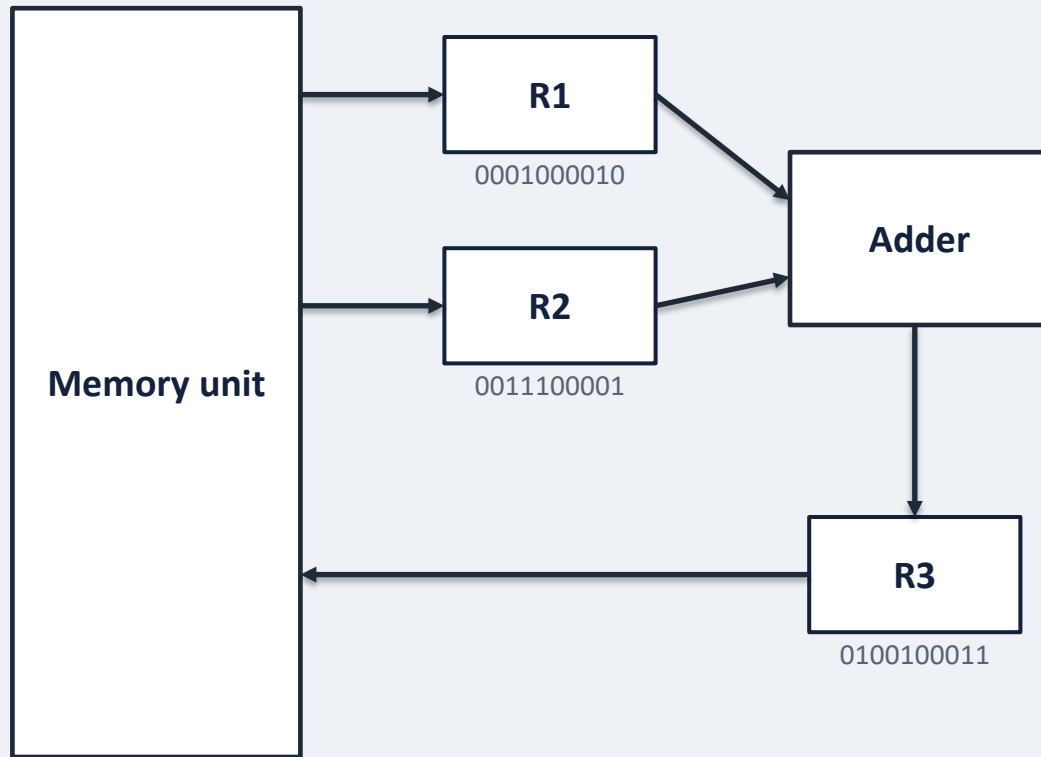
10 minutes — we resume at the 1:05 mark

Next: binary storage, registers and binary logic

Binary storage and registers

§1.8

Register transfer during an addition



A register

n binary cells store one n -bit word: 2^n states.

What does 1100 0011 1100 1001 mean?

Binary integer: 50,121

Two ASCII characters with even parity: 'C' 'I'

The code is an interpretation, not a property of the register.

Binary logic: AND, OR, NOT

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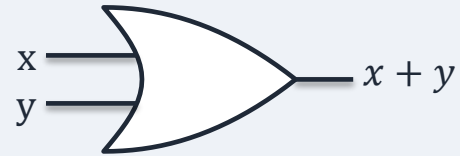
AND



x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

1 only when both are 1

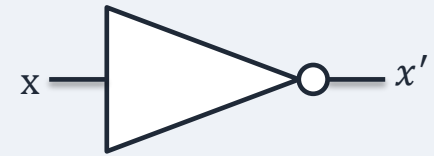
OR



x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

0 only when both are 0

NOT



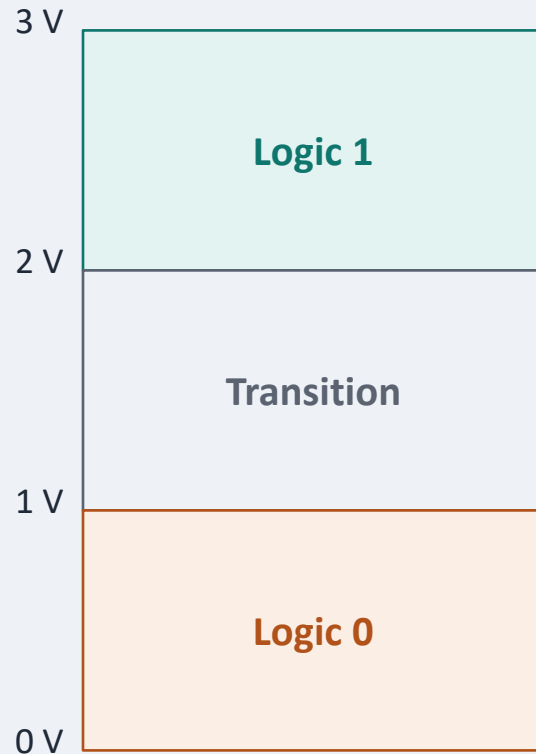
x	x'
0	1
1	0

the complement

Signal levels and multi-input gates

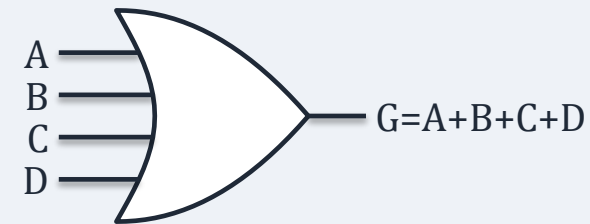
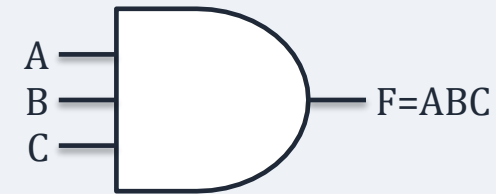
§2.8

Voltage ranges (example system)



A whole range counts as 1 or 0, so small noise does not change the value.

Gates with more inputs



AND: output 1 on exactly one of the 8 input rows.

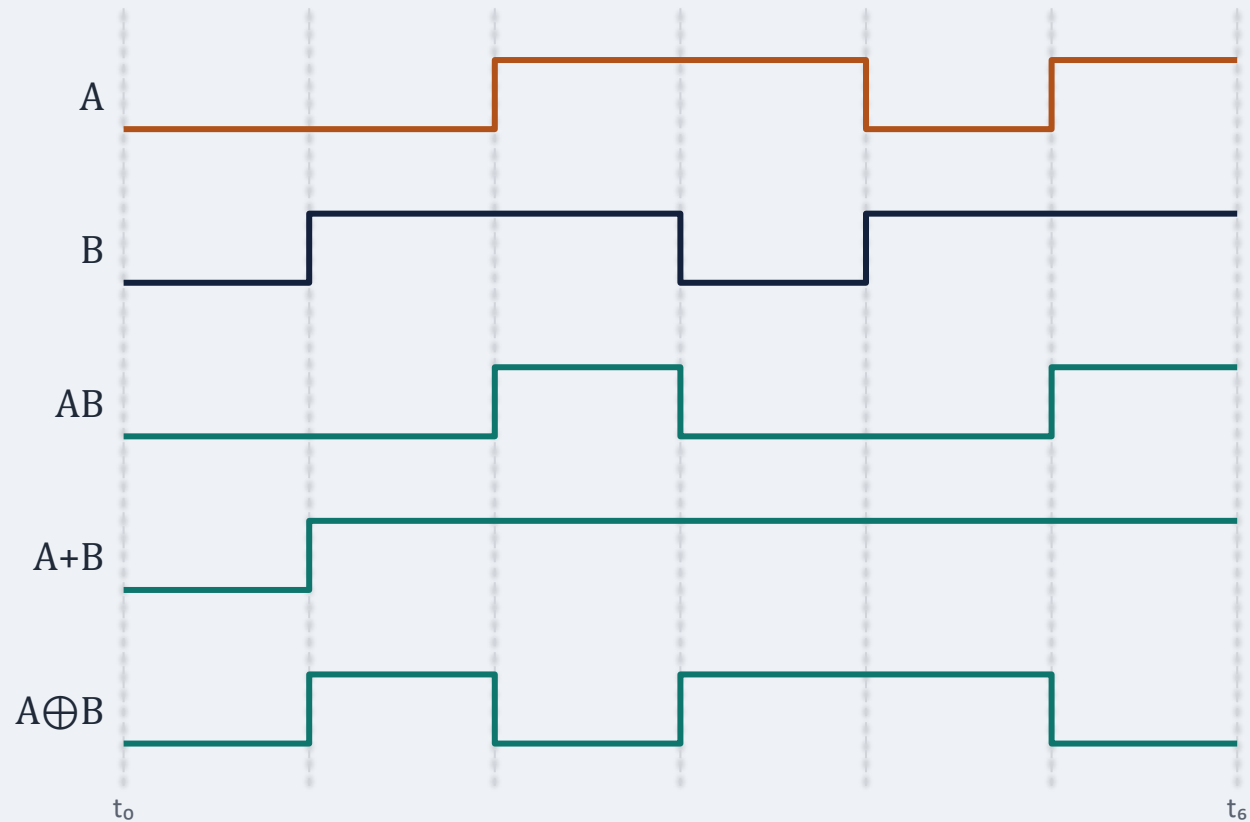
OR: output 0 on exactly one of the 16 input rows.

Example 4 — Reading a timing diagram

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EXAMPLE

A and B drive an AND, an OR and an XOR gate (no delay). Draw the three outputs.



- 1 Read one interval (column) at a time.
- 2 Write the inputs: $A = 0\ 0\ 1\ 1\ 0\ 1$, $B = 0\ 1\ 1\ 0\ 1\ 1$
- 3 AND = 1 only where both are 1: $0\ 0\ 1\ 0\ 0\ 1$
- 4 OR = 0 only where both are 0: $0\ 1\ 1\ 1\ 1\ 1$
- 5 XOR = 1 where they differ: $0\ 1\ 0\ 1\ 1\ 0$
- 6 With delay t_{pd} , each edge shifts right by t_{pd} .

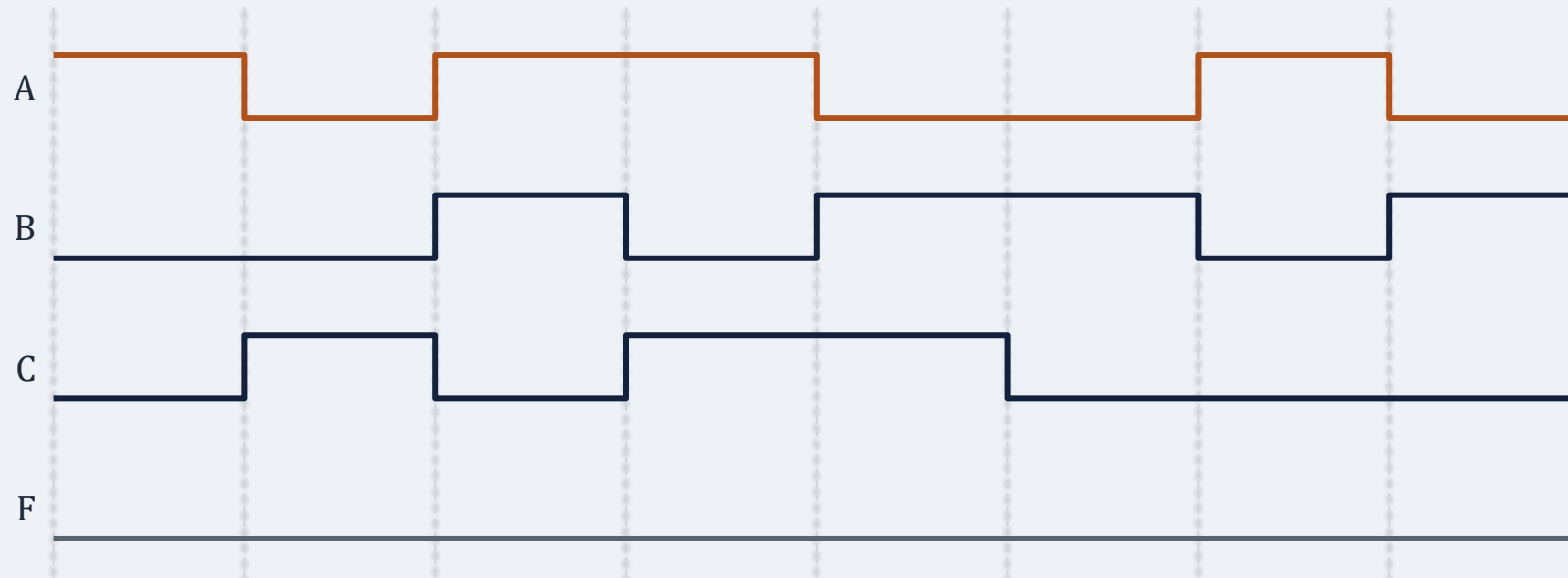
Your turn 2 — Draw the output

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YOUR TURN

≈ 8 min

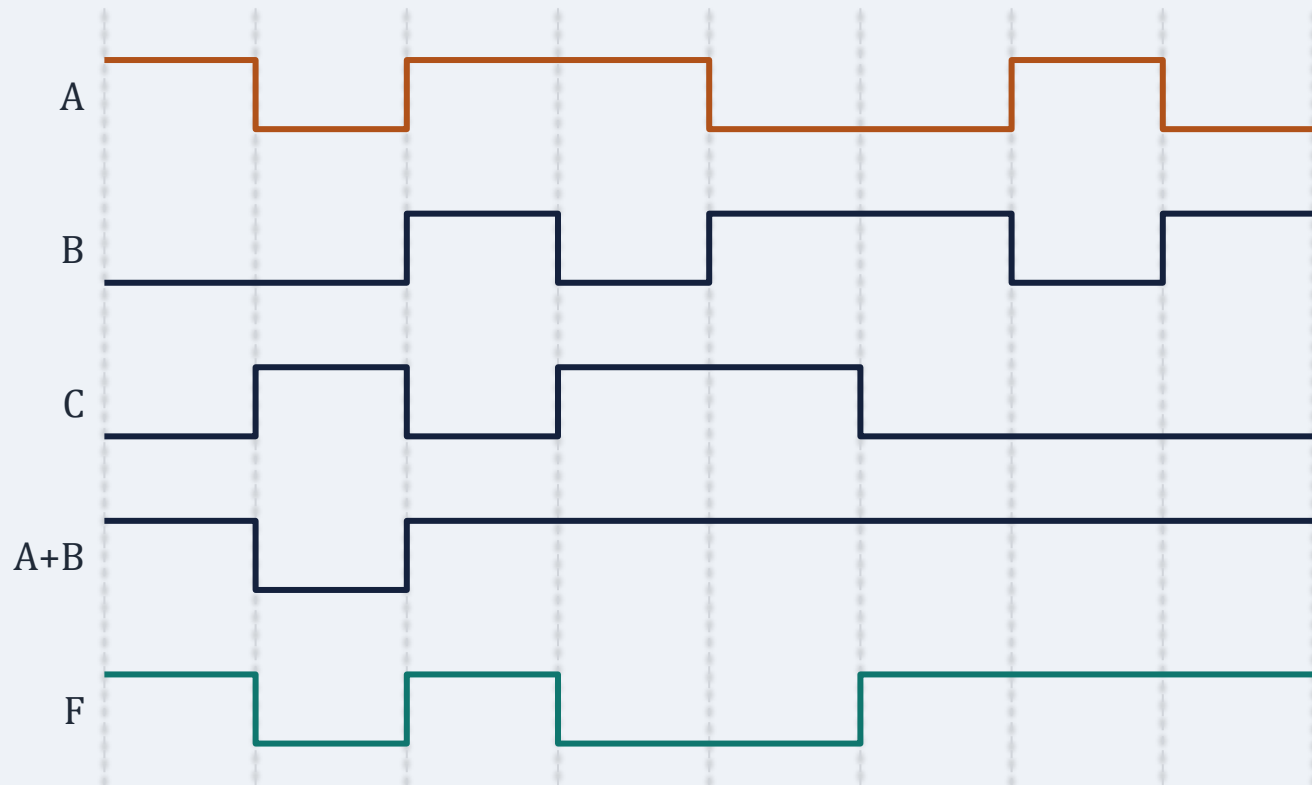
Draw $F = (A + B)C'$ for the inputs below (no gate delay).



Your turn 2 — Solution

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SOLUTION



- 1 Write the inputs by interval: $A = 1\ 0\ 1\ 1\ 0\ 0\ 1\ 0$
- 2 $B = 0\ 0\ 1\ 0\ 1\ 1\ 0\ 1$, $C = 0\ 1\ 0\ 1\ 1\ 0\ 0\ 0$
- 3 OR A and B : $A + B = 1\ 0\ 1\ 1\ 1\ 1\ 1\ 1$
- 4 Invert C : $C' = 1\ 0\ 1\ 0\ 0\ 1\ 1\ 1$
- 5 AND $A + B$ with C' , column by column.
- 6 $F = 1\ 0\ 1\ 0\ 0\ 1\ 1\ 1$

Concept check

1

Why does BCD need more bits than binary for the same number?

2

Why is Gray code used to label shaft encoders and K-maps?

3

A received word with even parity has four 1s. Is it correct?

4

Is $1 + 1 = 1$ or 10 ? Explain in logic and in arithmetic.

Key takeaways

BCD

4 bits per digit

Excess-3

BCD + 0011

Gray code

$$g_i = b_{i+1} \oplus b_i$$

Parity

detects odd # of errors

Register

n bits, 2^n states

Logic

AND · OR + NOT '

A bit pattern means what its code says it means.

Homework #1 due at the start of Lecture 3

This afternoon's lab: Lab X — Electronics basics

Next: Boolean algebra: postulates and theorems